

HW2 - Design Areas

Name: Sohum Sukhatankar

Collaborators: N/A

Idea

Personalized AI Study Companion - a mobile and web app powered by generative AI. It helps students learn more effectively through creating dynamic study plans, adaptive quizzes, AI tutoring sessions, and real-time progress feedback. Unlike static flashcards or generic video platforms, this tool tailors itself to each student's learning style and goals.

Similar Existing Services and Differentiation

- **Quizlet** - flashcards, but largely user-generated and non-adaptive.
- **Khan Academy** - excellent structured lessons but not personalized per learner.
- **ChatGPT** - can answer questions, but lacks progress tracking and targeted study flows.

Differentiation: This service will merge real-time generative AI with adaptive learning. It continuously evaluates what the student knows, generates tailored content, and updates study strategies. This only recently became feasible because of advances in generative AI models that can create high-quality, context-specific learning content at scale.

Stakeholders

- **Students (primary users):** Want effective, personalized study help. Motivated by academic performance and time efficiency.
- **Teachers/Educators:** Want a supplemental teaching tool but may want oversight of generated content. Motivated by student success.
- **Parents:** Want reliable tools that improve their children's learning outcomes. Concerned with cost, privacy, and safety.
- **Investors:** See opportunity in edtech market growth. Motivated by revenue, uniqueness, and scalability.

- **Regulators:** Care about data privacy (especially that of minors), algorithmic transparency, and ethical use of AI.
 - **Developers/Engineers:** Focus on creating robust, accurate, and user-friendly systems.
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User Persona & Journey

Persona: Jack, the Student

- **Backstory:** A 19-year-old college freshman majoring in biology. Struggles with keeping up in organic chemistry. Balances classes, part-time work, and extracurriculars.
 - **Values:** Efficiency, clarity, and confidence in mastering material.
 - **Motivations:** Wants to improve grades without burning out.
 - **Goal:** “I want a study system that adapts to me, so I don’t waste time studying the wrong way.”
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Current Journey (Without the AI Companion)

1. Jack logs into multiple resources (Canvas, Quizlet, YouTube) to find study material.
2. He spends 20 minutes searching for good explanations of a confusing concept.
3. He manually makes flashcards but struggles to know which areas to focus on.
4. After studying for hours, he still feels unsure about what he actually knows.
5. Before the test, he crams, repeats content he already mastered, and misses weak spots.

Pain Points:

- Too much time wasted finding and curating resources.
 - Anxiety about not knowing if he’s truly prepared.
 - Generic content that doesn’t fit his learning gaps.
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Improved Journey (With the AI Companion)

1. Jack logs into the AI Study Companion with his student account.
2. The system authenticates him and asks about his courses and upcoming exams.
3. He selects the folder for the course “Organic Chemistry”, uploads the relevant notes or materials (if any), and specifies: “Exam in 10 days, I struggle with resonance structures.”
4. The AI generates a personalized study plan with daily review chunks and quizzes.

5. Jack starts a study session. The AI tracks his responses, identifies that he's strong on mechanisms but weak on electron movement, and dynamically adjusts the material.
6. The AI tutor simulates an exam-style Q&A session, highlighting the specific regions where he struggles.
7. The app provides spaced repetition reminders and confidence metrics ("You've improved 40% in resonance problems since yesterday").
8. Jack feels confident and balanced, and he no longer has to cram.

The improvements include

- **Efficiency:** Jack no longer wastes time searching for study material; the app curates it instantly.
- **Personalization:** The AI detects his weak areas and tailors quizzes/explanations to those gaps.
- **Clarity:** Real-time feedback shows his progress, reducing test anxiety.
- **Retention:** Adaptive spaced repetition reinforces long-term memory.
- **Motivation:** Progress tracking and confidence boosts keep him engaged.

Values Tensions in Jack's Journey

1. Jack (Student) vs. Teachers/Educators

- Jack's value: Wants quick, adaptive help tailored to his weaknesses, even if explanations are simplified.
- Teacher's value: Wants rigor and alignment with course objectives, worried the AI may "oversimplify" or misrepresent difficult concepts.
- **Tension:** The AI must balance personalization with academic integrity and fidelity to the teacher's curriculum.

2. Jack vs. Parents

- Jack's value: Wants autonomy and privacy; may not want parents to see all his struggles.
- Parent's value: Want visibility into their child's learning progress and assurance the app is safe and worth paying for.
- **Tension:** Parents may push for monitoring dashboards, while Jack may resist oversight to preserve independence.

3. Jack vs. Regulators

- Jack's value: Wants maximum personalization, even if it means the system collects lots of personal data.
 - Regulator's value: Prioritize data minimization and compliance (FERPA, GDPR, COPPA if under 18).
 - **Tension:** Regulators may restrict how much adaptive tracking is allowed, potentially limiting the AI's ability to tailor study paths.
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4. Jack vs. Investors

- Jack's value: Wants affordability and accessibility (ideally free or low-cost).
 - Investor's value: Motivated by maximizing revenue, possibly via subscriptions, ads, or premium features.
 - **Tension:** The push for monetization may make the tool less accessible to lower-income students.
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5. Jack vs. Developers/Engineers

- Jack's value: Wants flawless, simple, and immediate performance.
- Developer's value: Must balance innovation with technical constraints (model cost, compute limits, bugs).
- **Tension:** Features Jack desires (e.g., real-time AI tutoring) may be difficult to deliver consistently, leading to frustration.